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COMICS

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— Linear Algebra & Stuffs —

a.) Diberikan $z_1 = 1+i$ dan $z_2 = \sqrt{3}-i$

I. Hitung z_1+z_2, z_1z_2, z_1^* dan z_1/z_2!

Jawab: ⇒ $z_1+z_2 = (1+i)+(\sqrt{3}-i) = \boxed{1+\sqrt{3}}$ //

⇒ $z_1z_2 = (1+i)(\sqrt{3}-i) = \sqrt{3}-i+i\sqrt{3}-i^2$

$= \boxed{(\sqrt{3}+1)+(\sqrt{3}-1)i}$ //

⇒ $z_1^* = \boxed{1-i}$ //

⇒ $z_1/z_2 = 1+i/\sqrt{3}-i = (1+i)(\sqrt{3}+i)/(\sqrt{3}-i)(\sqrt{3}+i)$

$= \frac{\sqrt{3}+i+i\sqrt{3}-1}{3-(-1)} = \boxed{\frac{(\sqrt{3}-1)+(\sqrt{3}+1)i}{4}}$ //

II. Hitung $|z_1|$, $|z_2|$, dan $|z_1z_2|$ serta verifikasi $|z_1z_2| = |z_1||z_2|$

Jawab: ⇒ $|z_1| = \sqrt{1^2+1^2} = \boxed{\sqrt{2}}$ //

⇒ $|z_2| = \sqrt{(\sqrt{3})^2+(-1)^2} = \sqrt{3+1} = \sqrt{4} = \boxed{2}$ //

⇒ $|z_1z_2| = \sqrt{(\sqrt{3}+1)^2+(\sqrt{3}-1)^2} = \sqrt{8} = \boxed{2\sqrt{2}}$ //

(Terverifikasi $|z_1||z_2| = \sqrt{2}\cdot 2 = 2\sqrt{2} = |z_1z_2|$) //

III. Tuliskan z_1 dan z_2 dalam bentuk polar $re^{i\theta}$

Jawab: ⇒ $z_1 = \boxed{\sqrt{2}e^{i\pi/4}}$ // ($r = \sqrt{2}, \theta = \arctan(1/1) = \pi/4$)

⇒ $z_2 = \boxed{2e^{-i\pi/6}}$ // ($r = 2, \theta = \arctan(-1/\sqrt{3}) = -\pi/6$)

IV. Gunakan bentuk polar untuk menghitung z_1^8!

Jawab: $z_1^8 = (\sqrt{2}e^{i\pi/4})^8 = (\sqrt{2})^8 e^{i(8\pi/4)} = 16e^{i2\pi} = 16(\cos 2\pi + i\sin 2\pi)$

$= \boxed{16}$ //

V. Tunjukkan bahwa $|e^{i\theta}|^2 = 1$ untuk sembarang θ real:

Jawab: $e^{i\theta} = \cos\theta + i\sin\theta \rightarrow |e^{i\theta}| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$

$|e^{i\theta}|^2 = 1^2 = \boxed{1}$ //

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02.) Diberikan state (belum tentu ternormalisasikan)

$|a\rangle = (1\ 2i)$, $|b\rangle = (2\ -i)$, $|c\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2} e^{i\pi/4}|1\rangle$

I. Tuliskan $\langle a|$ dan $\langle b|$

Jawab: ⇒ $\langle a| = (1\ -2i)$

⇒ $\langle b| = (2\ \ i)$ //

II. Hitung $\langle a|b\rangle$ dan $\langle b|a\rangle$ serta verifikasi $\langle a|b\rangle = \langle b|a\rangle^*$:

Jawab: ⇒ $\langle a|b\rangle = (1)(2) + (-2i)(-i) = 2 - 2 = 0$ //

⇒ $\langle b|a\rangle = (2)(1) + (i)(2i) = 2 - 2 = 0$

$\langle b|a\rangle^* = 0^* = 0 = \langle a|b\rangle$ //

III. Apakah $|a\rangle$ dan $|b\rangle$ ortogonal?

Jawab: Ya, karena $\langle a|b\rangle = 0$ //

IV. Normalisasi $|a\rangle$ dan $|b\rangle$!

Jawab: - $\| \langle a|a\rangle \|^{1/2} = \sqrt{1^2 + |2i|^2} = \sqrt{5} \rightarrow |a'\rangle = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ 2i \end{pmatrix}$ //

- $\| \langle b|b\rangle \|^{1/2} = \sqrt{2^2 + |-i|^2} = \sqrt{5} \Rightarrow |b'\rangle = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ -i \end{pmatrix}$ //

V. Apakah $|c\rangle$ sudah ternormalisasikan? Hitung $P(0)$ dan $P(1)$!

Jawab. $|1/2|^2 + |\sqrt{3}/2\, e^{i\pi/4}|^2 = 1/4 + 3/4 = 1$ (Sudah) //

⇒ $P(0) = |\langle 0|c\rangle|^2 = (1/2)^2 = 1/4 = 0{,}25$ //

⇒ $P(1) = |\langle 1|c\rangle|^2 = (\sqrt{3}/2)^2 = 3/4 = 0{,}75$ //

VI. Tuliskan $|c\rangle$ dalam Hadamard basis $|+\rangle, |-\rangle$, lalu hitung $P(+)$ & $P(-)$

Jawab: $|c\rangle = \frac{1}{2\sqrt{2}} (1 + \sqrt{3} e^{i\pi/4}) |+\rangle + \frac{1}{2\sqrt{2}} (1 - \sqrt{3} e^{i\pi/4}) |-\rangle$

Dengan $e^{i\pi/4} = 1+i/\sqrt{2}$, diperoleh $P(+) = 4+\sqrt{6}/8 \approx 0{,}8003$ dan $P(-) = 4-\sqrt{6}/8 \approx 0{,}1997$ //

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VII. Buktikan $|+\rangle, |-\rangle$ adalah basis ortonormal!
Jawab: $\langle +|+\rangle = 1$, $\langle -|-\rangle = 1$, $\langle +|-\rangle = 0$ //

03.) Diberikan matriks:

$$A = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

I. Hitung $A^\dagger$, $S^\dagger$, $B^\dagger$!
Jawab: $A^\dagger = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix}$, $S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$, $B^\dagger = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ //

II. Tentukan mana yang Hermitian, mana yang unitary (boleh keduanya / tidak sama sekali).
Jawab: - A (Hermitian) ($A^\dagger = A$), bukan unitary
- S (Unitary) ($S^\dagger S = I$), bukan Hermitian
- B (Bukan Hermitian dan Unitary) ($B^\dagger B \neq I$) //

III. Hitung $S|+\rangle$. State apa yang dihasilkan!?
Jawab: $S|+\rangle = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$ // (eigen vector) //

IV. Hitung HZH dan HXH dengan H adalah Hadamard. Apa yang Anda amati?
Jawab. $HZH = X$. dan $HXH = Z$ (gerbang Hadamard merubah sumbu X dan Z). //

V. Hitung Commutator $[X, Y]$:
Jawab: $[X, Y] = XY - YX = \boxed{2iZ}$ //

VI. Tunjukkan $X^2 = Y^2 = Z^2 = I$!
Jawab: Perkalian matriks Pauli menghasilkan dirinya sendiri (I). //

VII. Mengapa B bukan quantum gate?
Jawab: B tidak unitary sehingga mengubah norma/panjang state.

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04.)

I. Hitung eigenvalue dan eigenvector ternormalisasi dari Pauli Y secara manual ?

Jawab : → Eigenvalue $(\lambda) = 1, -1$
→ Eigenvector : $|v_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} -i \\ 1 \end{pmatrix}$ dan $|v_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} i \\ 1 \end{pmatrix}$ //

II. Hitung eigenvalue dan eigenvector matriks A:

Jawab: → Eigenvalue: $\lambda_1 = 1$ //
$\lambda_2 = 4$

→ Eigenvector: $|u_1\rangle = \frac{1}{\sqrt{3}}\begin{pmatrix} i \\ -1 \end{pmatrix}$ $A|u_2\rangle = \frac{1}{\sqrt{6}}\begin{pmatrix} 1-i \\ 2 \end{pmatrix}$ //

III. Verifikasi ortogonalitas eigenvector A:

Jawab: $\langle u_1 | u_2 \rangle = 0$ (ortogonal) //

IV. Eigenvalue Hadamard H!

Jawab: $\lambda = 1$ dan $\lambda = -1$

V. Spectral decomposition Pauli Z!

Jawab: $Z = (1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1| = |0\rangle\langle 0| - |1\rangle\langle 1|$ //

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